

Breaking the Fixed-Sentinel Blind Spot in Formation-Preserving GNSS Spoofing: Randomized Anchor Challenges and a Swarm Rigid-Transform Protection Level

Utsav Boricha, Harshil Patel, Krishiv Patel, Hrishi Shah, and Yash Darji

Department of Computer Science and Engineering

Indian Institute of Information Technology Vadodara – International Campus Diu
Diu, India

Abstract—Formation-preserving global navigation satellite system (GNSS) spoofing can move the absolute position reported by every unmanned aerial vehicle (UAV) while leaving inter-UAV distances unchanged. We show a stricter blind spot: for any selected sentinel, a two-dimensional rigid attack can rotate the remaining swarm about that sentinel, leaving both all pairwise distances and the sentinel’s absolute position unchanged. Consequently, neither relative ranging nor a permanently audited leader is sufficient. We propose randomized authenticated anchor challenges that unpredictably select a swarm member and a subset of ground anchors. Challenge residuals provide detection, while their geometry estimates a three-parameter rigid error state. From its information matrix we define a Swarm Rigid-Transform Protection Level (SRPL), which is infinite when absolute translation and yaw are unobservable and otherwise upper-bounds the model-based horizontal displacement of every member. A weighted Procrustes step then provides a swarm-wide correction when the challenge geometry is adequate. In seeded position-level Monte Carlo experiments with five UAVs, four anchors, 30% packet loss, and 500 runs per detector/scenario cell, the proposed audit detected 100% of rotations about the fixed leader; relative-only and fixed-leader checks detected 0%. The nominal run-level alarm rate was 0.6%. A schedule using three anchor ranges per second reduced ranging activity by 96% relative to continuous three-anchor polling of all members. All 300 protection-level runs obtained a finite SRPL, with a median first availability of 2.6 s and a median 5-s SRPL of 7.95 m. Recovery experiments achieved a median swarm root-mean-square error of 2.33 m. These are reproducible simulation results, not flight certification; hardware-in-the-loop and licensed shielded testing remain necessary.

Index Terms—GNSS spoofing, UAV swarm, formation-preserving attack, UWB ranging, integrity monitoring, protection level, randomized audit

I. INTRODUCTION

GNSS receivers provide globally available position, velocity, and time, but civil signals can be counterfeited or manipulated [1], [2]. Controlled demonstrations have shown that spoofed navigation can capture an unmanned aircraft’s guidance loop [3]. A swarm appears to offer redundancy: its members can compare GNSS-derived separations with inter-UAV ranging. That intuition fails under a coherent spatial attack. Ceccato *et al.* analyzed spatial spoofing against

multi-antenna drone swarms [4]; more recently, GNSS-WASP demonstrated wide-area manipulation that preserves a victim formation within legitimate noise [5]. Such an attack can bypass detectors that only compare relative distances.

One response is to add an absolute trusted receiver or anchor. This paper identifies why a fixed sentinel is still insufficient. A rotation about that sentinel leaves its reported location exactly correct while displacing every other member. Randomness is therefore not merely a resource-saving scheduler; it removes a deterministic geometric blind spot by preventing the protected member from being known in advance.

We propose Randomized Authenticated Challenge with a Swarm Rigid-Transform Protection Level (RAC-SRPL). The design uses sparse authenticated ultra-wideband (UWB) anchor ranging as a diverse absolute reference. It does not assume that UWB is intrinsically secure: IEEE 802.15.4z time-of-flight ranging has known implementation and physical-layer limitations [6], [7], and the system must use a reviewed secure-ranging profile such as the mechanisms studied in [8]. The contributions are:

- a proof that pairwise checks and any permanently audited single member share a rigid-transform blind spot;
- an unpredictable member-and-anchor challenge protocol, with a closed-form scheduling relation between challenge rate and detection probability;
- an absolute SRPL that reports infinity when the swarm’s translation and yaw error are not observable, instead of returning a misleading finite relative bound;
- a correction policy based on rigid registration, together with a reproducible seeded Monte Carlo evaluation across local, translational, rotational, and slow-drift cases; and
- an explicit outcome envelope that distinguishes continued operation, recovery, and fail-safe behavior.

Our bounded novelty claim is as follows: to the best of our literature search completed on Aug. 7, 2026, we found no prior work that jointly treats a formation-preserving wide-area threat, randomizes authenticated absolute challenges across

swarm members, and uses the resulting geometry to compute an absolute rigid-transform protection bound that gates swarm-wide recovery. This claim is deliberately narrower than saying that UWB spoof detection, cooperative integrity, spatial spoofing, or robust swarm localization is itself new.

II. RELATED WORK AND NOVELTY POSITION

Receiver-level GNSS spoofing detection includes signal-quality, angle-of-arrival, multi-antenna, inertial, and cryptographic methods [1]. These are complementary to the geometric method proposed here. Galileo Open Service Navigation Message Authentication (OSNMA), operational since July 2025, authenticates navigation data, but EUSPA notes that pseudorange-level signal authentication is a later complementary service [9]. Thus, navigation-message authentication is valuable defense in depth but does not by itself close every ranging-level manipulation path.

For multiple UAVs, Mykityn *et al.* compare GPS and pairwise UWB distances to detect inconsistent members [10]. Sun fuses double-differenced GNSS pseudoranges and UWB to calculate a *relative* protection level for multi-UAV navigation [11]. Relative integrity is essential for collision avoidance, but a rigid displacement is in the null space of all pairwise ranges. Dang *et al.* use a graph neural network and cellular information for swarm detection [12], while Michieletto *et al.* combine inertial and relative ranging for robust formation localization [13]. These approaches motivate cooperative diversity, but they do not expose uncertainty specifically as an absolute swarm rigid-transform bound.

Table I states the distinction without claiming that adjacent components are absent from prior work.

III. SYSTEM AND THREAT MODEL

Consider $N \geq 2$ UAVs flying in a horizontal plane. The true position of member i is $\mathbf{p}_i \in \mathbb{R}^2$ and its GNSS position output is $\mathbf{z}_i^G$. The swarm also measures selected inter-UAV ranges and can communicate over an authenticated mesh. A protected operating region contains $L \geq 3$ surveyed ground anchors at $\mathbf{a}_\ell \in \mathbb{R}^2$. An anchor can perform challenge-response time-of-flight ranging with a selected UAV.

The attacker may coherently alter the GNSS position output of several or all members. The strongest modeled common-mode component is a two-dimensional rigid transform

$$\mathbf{z}_i^G = \mathbf{R}(\psi)\mathbf{p}_i + \mathbf{t} + \mathbf{v}_i^G, \quad (1)$$

where $\mathbf{R}(\psi) \in \text{SO}(2)$, $\mathbf{t} \in \mathbb{R}^2$, and $\mathbf{v}_i^G$ is receiver noise. Local faults and slowly changing transforms are also evaluated. The attacker may observe ordinary network traffic but cannot predict fresh cryptographic nonces before a challenge, compromise trusted anchor keys, or arbitrarily reduce a securely verified distance without being detected. Packet loss and missing challenges are allowed. Denial of service remains possible and must cause loss of integrity availability, not silent acceptance.

We do not model compromise of the autopilot, anchor coordinates, or all UWB hardware. We also do not claim that the

Gaussian covariance model covers multipath, non-line-of-sight bias, or adversarial UWB implementation flaws. Those factors require fault hypotheses and hardware calibration before safety claims.

IV. TWO GEOMETRIC BLIND SPOTS

Let a noiseless pairwise range be $d_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\|_2$.

A. Pairwise invariance

Under the rigid component of (1),

$$\begin{aligned} \|\mathbf{z}_i^G - \mathbf{z}_j^G\|_2 &= \|\mathbf{R}(\psi)(\mathbf{p}_i - \mathbf{p}_j)\|_2 \\ &= \|\mathbf{p}_i - \mathbf{p}_j\|_2. \end{aligned} \quad (2)$$

Equation (2) proves that every pairwise-distance residual can remain nominal during arbitrary common translation and rotation. More pairwise links improve local-fault detection but do not recover absolute observability.

B. Fixed-sentinel invariance

Suppose member s is always the sole absolute sentinel. For any chosen $\mathbf{R} \neq \mathbf{I}$, set

$$\mathbf{t} = (\mathbf{I} - \mathbf{R})\mathbf{p}_s. \quad (3)$$

Then

$$\mathbf{z}_s^G = \mathbf{p}_s, \quad \mathbf{z}_i^G = \mathbf{p}_s + \mathbf{R}(\mathbf{p}_i - \mathbf{p}_s), \quad (4)$$

so the sentinel and all inter-UAV ranges are unchanged while, in general, every $i \neq s$ is displaced. Figure 1 shows the evaluated 18° case. This construction is the fixed-sentinel blind spot.

V. RANDOMIZED AUTHENTICATED ANCHOR CHALLENGES

A. Challenge and residual test

At each navigation epoch k , the controller draws $B_k \sim \text{Bernoulli}(q)$. If $B_k = 1$, it chooses member I_k uniformly at random and selects at least three non-collinear anchors. A challenge includes fresh nonces, member and anchor identities, epoch binding, and authenticated response data. The protocol obtains ranges

$$\rho_{i\ell} = \|\mathbf{p}_i - \mathbf{a}_\ell\|_2 + \eta_{i\ell}. \quad (5)$$

The anchor-local position is the weighted nonlinear least-squares solution

$$\hat{\mathbf{p}}_i^A = \arg \min_{\mathbf{x} \in \mathbb{R}^2} \sum_{\ell \in \mathcal{L}_k} \frac{(\rho_{i\ell} - \|\mathbf{x} - \mathbf{a}_\ell\|_2)^2}{\sigma_{i\ell}^2}. \quad (6)$$

Define $\mathbf{y}_i = \mathbf{z}_i^G - \hat{\mathbf{p}}_i^A$ and $\mathbf{C}_i = \mathbf{P}_i^G + \mathbf{P}_i^A$. The normalized innovation statistic is

$$T_i = \mathbf{y}_i^T \mathbf{C}_i^{-1} \mathbf{y}_i. \quad (7)$$

Under a calibrated zero-mean Gaussian null, $T_i \sim \chi_2^2$; an audit alarms when $T_i > \chi_{2,1-\alpha}^2$.

If m of N members have a detectable bias, packet-loss probability is p_{loss} , and an audit of a biased member exceeds the threshold with probability P_D , the per-epoch alarm probability is

$$\pi = q(1 - p_{\text{loss}}) \frac{m}{N} P_D. \quad (8)$$

TABLE I
POSITION OF THE PROPOSED WORK RELATIVE TO THE CLOSEST RESEARCH THEMES

Work or theme	Primary protected/analyzed quantity	Absolute-reference scheduling	Integrity or recovery output
Sun [11]	Relative multi-UAV position using GNSS and UWB	No randomized member challenge	Relative protection level
Mykityn <i>et al.</i> [10]	Consistency of pairwise GPS and UWB separations	No randomized absolute audit	Spoofing alarm and identification logic
Ceccato <i>et al.</i> [4]; GNSS-WASP [5]	Spatial/formation-preserving attack feasibility	Attack and countermeasure analysis	Attack error or detection discussion
Michieletto <i>et al.</i> [13]	Robust formation localization using inertial and relative measurements	No randomized anchor challenge	Secure position estimate
This work	Absolute translation and yaw error shared by a swarm	Random member and anchor subset, authenticated and nonce-bound	Alarm, finite/infinite SRPL, and gated rigid correction

Assuming independent scheduling, the probability of at least one alarm within H epochs is

$$P_{\text{alarm}}(H) = 1 - (1 - \pi)^H. \quad (9)$$

Equation (9) gives a direct way to choose q from a detection-time target. It also shows why a permanently selected member is qualitatively different from random selection: in (4), the fixed member has $P_D \approx \alpha$ while $m = N - 1$ under randomized selection.

B. Rigid-error information and SRPL

Detection alone does not tell the controller whether enough geometry exists to bound every member's absolute error. Let the small residual rigid error about the current estimate be

$$\xi = [t_x \quad t_y \quad \delta\psi]^T. \quad (10)$$

For centered member coordinate $\tilde{\mathbf{p}}_i = \mathbf{p}_i - \mathbf{p}_c$ and $\mathbf{J} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, linearization gives

$$\mathbf{y}_i \approx \mathbf{G}_i \xi + \epsilon_i, \quad \mathbf{G}_i = [\mathbf{I}_2 \quad \mathbf{J}\tilde{\mathbf{p}}_i]. \quad (11)$$

For a recent challenge window $\mathcal{A}_k$, retaining at most the latest audit of each distinct member, define

$$\Lambda_k = \sum_{i \in \mathcal{A}_k} \mathbf{G}_i^T \mathbf{C}_i^{-1} \mathbf{G}_i. \quad (12)$$

One unique audited member contributes only two independent equations for the three unknown transform parameters, so yaw remains unobservable. In two dimensions, two distinct audited members with nonzero separation are sufficient in the noise-free generic case. Repeated audits of one sentinel must not be counted as geometric diversity.

For integrity multiplier $\gamma = \chi_{3,1-\beta}^2$, our model-based Swarm Rigid-Transform Protection Level is

$$\text{SRPL}_k = \begin{cases} \max_j \sqrt{\gamma \lambda_{\max}(\mathbf{G}_j \Lambda_k^{-1} \mathbf{G}_j^T)}, & \text{rank}(\Lambda_k) = 3, \\ +\infty, & \text{otherwise.} \end{cases} \quad (13)$$

Unlike a relative protection level, (13) explicitly includes the common translation and yaw that move the formation in the

global frame. A mission may declare absolute navigation available only if SRPL_k is finite and below a horizontal alert limit H_{AL} . The confidence interpretation is conditional on the covariance and fault model; it is not a certified aviation integrity guarantee.

C. Swarm-wide correction

Once at least two distinct members have fresh anchor fixes and the geometry passes the SRPL gate, a rigid correction can be estimated through weighted orthogonal Procrustes registration [14]:

$$(\hat{\mathbf{R}}, \hat{\mathbf{t}}) = \arg \min_{\substack{\mathbf{R} \in \text{SO}(2) \\ \mathbf{t} \in \mathbb{R}^2}} \sum_{i \in \mathcal{A}_k} \|\hat{\mathbf{p}}_i^A - (\mathbf{R}\mathbf{z}_i^G + \mathbf{t})\|_{\mathbf{C}_i^{-1}}^2. \quad (14)$$

The controller forms $\tilde{\mathbf{p}}_j = \hat{\mathbf{R}}\mathbf{z}_j^G + \hat{\mathbf{t}}$ for every member. A correction is never silently applied when (13) is infinite, anchor residuals are mutually inconsistent, or the alert limit is exceeded.

VI. OUTCOME-GATED RESPONSE

Table II lays out the operational outcomes generated by the method. It is intentionally conservative: packet loss or weak geometry reduces availability instead of being interpreted as evidence that GNSS is safe.

VII. REPRODUCIBLE SIMULATION

A. Configuration

The evaluation is a position-level simulation, not an RF signal generator. Five UAVs follow a smooth formation trajectory for 30 s at 5 Hz. Four anchors surround the operating area. The assumed one-standard-deviation noises are 1.50 m per horizontal GNSS axis, 0.15 m for inter-UAV UWB range, and 0.20 m for anchor range. At every epoch, $q = 0.20$; a successful challenge audits one uniformly random UAV through three of four anchors. Independent challenge packet loss is 30%. The audit false-alarm allocation is $\alpha = 10^{-4}$, and the SRPL uses $\gamma = \chi_{3,0.9999}^2 = 21.11$. A 5-s window retains the latest challenge from each distinct member.

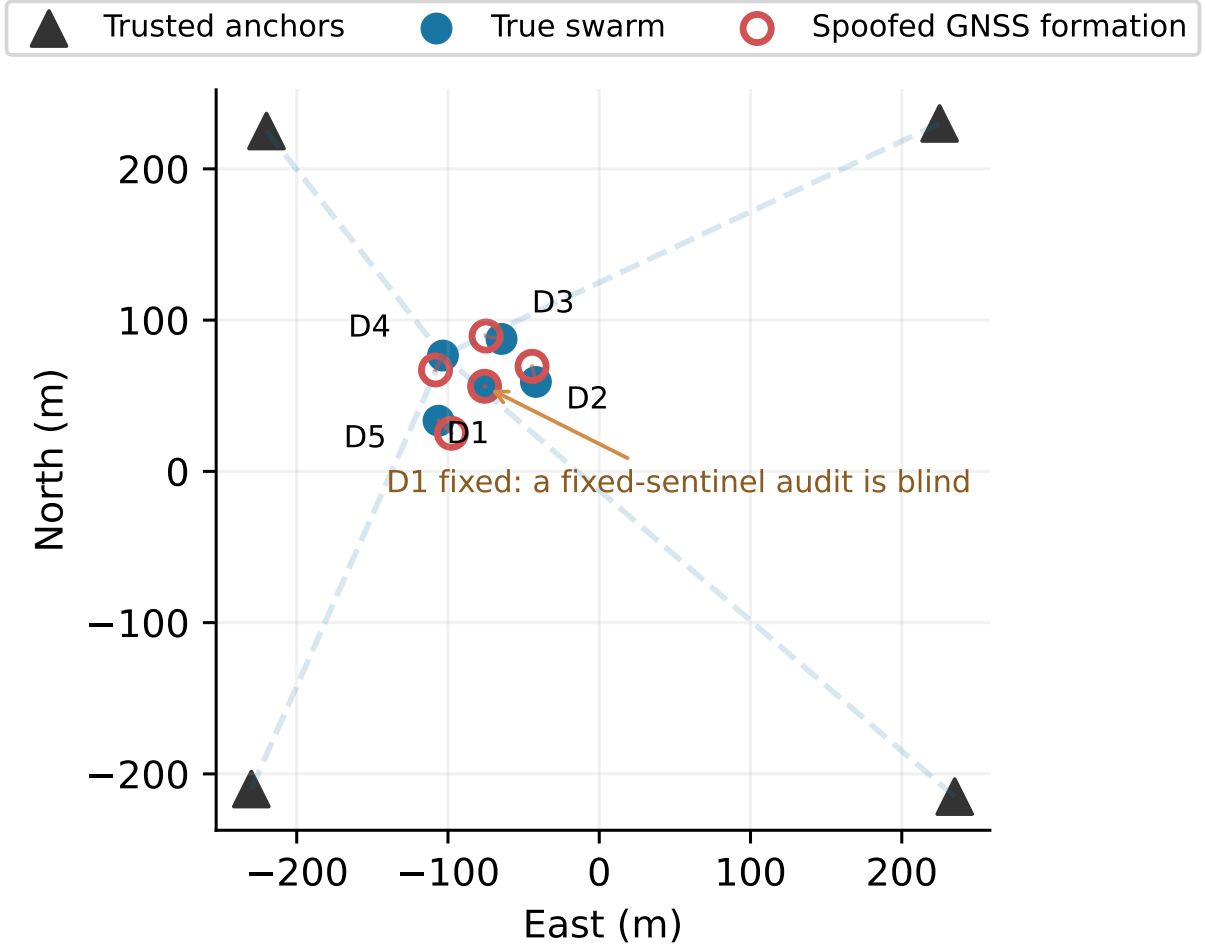


Fig. 1. Formation-preserving rotation about UAV D1. D1 and all pairwise distances are unchanged; an unpredictable challenge to another member supplies the missing absolute constraint. The diagram shows the defensive geometry evaluated in this paper.

We ran 500 independent trials for every detector/scenario cell. Three detectors were compared: (i) all inter-UAV ranges versus GNSS-derived ranges, using a five-standard-deviation maximum residual; (ii) the same challenge schedule but always auditing leader D1; and (iii) the proposed random-member schedule. Scenarios were nominal operation, a 25/-18 m offset on one UAV, a 28/22 m common translation, an 18° rotation about D1, and a slow 20-s combined translation/rotation drift. Separately, 300 trials per scheduling/geometry point evaluated SRPL availability, and 1000 trials per packet-loss value evaluated recovery from the D1-centered rotation. The recovery prototype uses equal-weight Procrustes, a special case of (14). All random streams use master seed 20260807.

B. Detection results

Table III and Fig. 2 give run-level outcomes. Relative ranging detected the single-member fault in all runs but, exactly as (2) predicts, detected none of the three common rigid cases. The fixed-leader anchor detected common translation and slow drift but none of the rotations constructed about

D1. Randomized challenges detected the D1-centered rotation in all 500 runs, with median delay 1.2 s and 95th-percentile delay 4.81 s. They also detected all common translations (0.8 s median) and slow drifts (5.8 s median), and 97.6% of local offsets (4.6 s median). Its nominal run-level alarm rate was 0.6% over 30 s.

For context, 500/500 corresponds to a 95% Wilson interval of 99.2–100%, 0/500 to 0–0.8%, and 488/500 to 95.9–98.6%. Thus, “100%” here means the observed finite Monte Carlo sample, not a universal detection guarantee.

C. Protection, overhead, and recovery

Continuous polling of all five members through three anchors at 5 Hz requires 75 anchor ranges/s. The proposed $q = 0.20$ schedule requires an expected three ranges/s, a 96% reduction. With surrounding anchors and 30% packet loss, every one of 300 runs eventually obtained a finite SRPL. Median time to first finite SRPL was 2.6 s; the median fraction of post-warm-up time with a finite bound was 85.6%, and the median 5-s SRPL was 7.95 m. Figure 3 exposes the

TABLE II
DECISION OUTCOMES FOR A DEFENSIVE PROTOTYPE

Anchor challenge	SRPL	Relative check	Permitted response
Nominal	Finite and $\leq H_{AL}$	Nominal	Continue with monitored GNSS; schedule the next unpredictable audit.
Alarm	Finite and $\leq H_{AL}$ after exclusion	Nominal	Reject suspect GNSS transform, compute rigid correction, and continue only within the validated operating envelope.
Alarm	Infinite or $> H_{AL}$	Any	Do not correct from insufficient geometry; hold, move to anchor/inertial navigation, return, or land according to mission safety logic.
Nominal anchor audit	Finite	Relative alarm	Isolate a local member/sensor fault; do not classify it automatically as a common rigid attack.
Missing, stale, or mutually inconsistent	Infinite by policy	Any	Treat the absolute reference as unavailable or attacked; enter fail-safe and log evidence.
Persistent OSNMA failure	Any	Any	Reject the affected GNSS data as a separate defense-in-depth alarm; never use a good geometry score to override failed authentication.

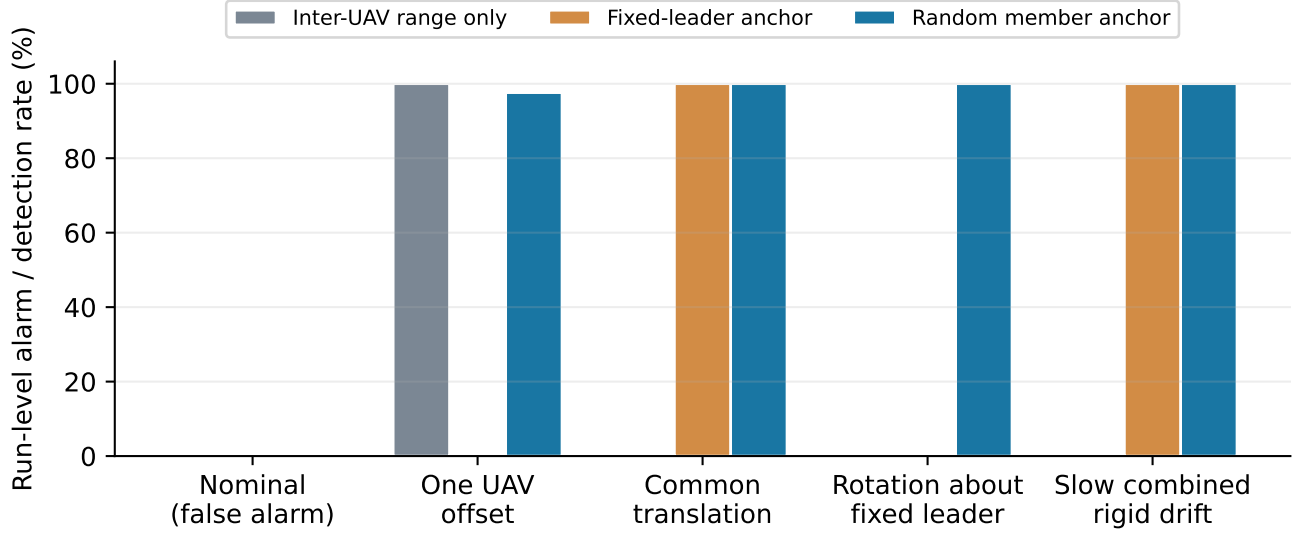


Fig. 2. Observed run-level alarm/detection rate. A common rigid transform is invisible to pairwise range checks; rotation about D1 is also invisible to a fixed-D1 absolute audit.

TABLE III
RUN-LEVEL ALARM/DETECTION RATE (%), 500 RUNS PER CELL

Detector	Nom.	Local	Trans.	Rot.-D1	Slow
Relative range only	0.0	100.0	0.0	0.0	0.0
Fixed-leader anchor	0.4	0.2	100.0	0.0	100.0
Random-member anchor	0.6	97.6	100.0	100.0	100.0

intended trade-off: more challenges improve bound size and availability but consume more channel/energy resources. One-sided anchor geometry produced larger bounds at matched settings, confirming that anchor placement matters even when rank is technically sufficient.

At 30% packet loss, all 1000 recovery trials acquired two distinct audited members within the 30-s horizon. Median acquisition time was 2.6 s (95th percentile 7.0 s). After correcting the 18° D1-centered attack, the median five-member position root-mean-square error (RMSE) was 2.33 m and the 95th percentile was 4.60 m. At 50% packet loss, the median and 95th-percentile acquisition times increased to 3.6 and 10.2 s, while median RMSE remained 2.36 m. The delay increase, rather than silent confidence, is the correct consequence of missing challenges.

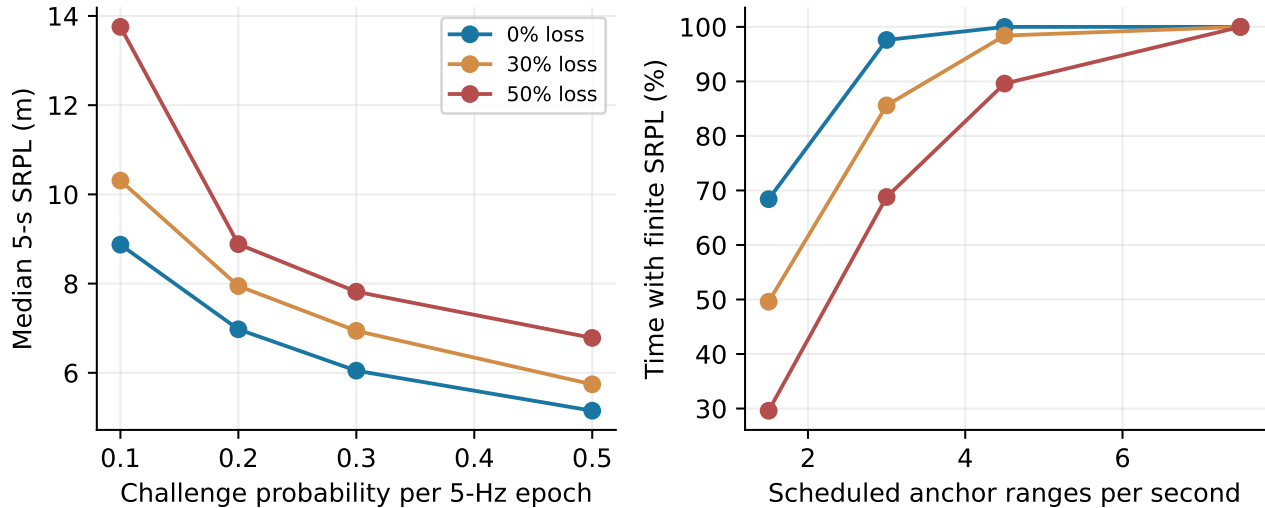


Fig. 3. Surrounding-anchor scheduling trade-off over 300 runs per point. “Finite” means that two or more distinct recent members made the three-parameter rigid state observable; it is not a certification claim.

VIII. LIMITATIONS AND VALIDATION PATH

The current evidence supports a research hypothesis, not operational deployment. First, the simulation injects position-domain rigid errors and Gaussian noise; it does not reproduce receiver tracking loops, multipath, non-line-of-sight UWB bias, correlated GNSS errors, anchor clock faults, or closed-loop vehicle dynamics. Second, (11) is a local approximation. Large transforms should be estimated by the nonlinear registration in (14), followed by re-linearization before reporting SRPL. Third, the present SRPL is covariance-based and conditional on the modeled fault set. A publication-quality integrity claim needs explicit missed-detection risk allocation, noncentral test distributions, calibration data, and overbounding distributions.

The next experimental stages are: (1) software-in-the-loop flight-control replay; (2) hardware-in-the-loop with recorded GNSS observables and real UWB tags/anchors; (3) shielded or cabled receiver testing; (4) benign outdoor flight to calibrate line-of-sight and non-line-of-sight residuals; and only then (5) a regulator-approved radiating trial, if scientifically necessary. In India, the Department of Telecommunications lists separate experimental licenses for non-radiating and radiating wireless trials [15]. No over-the-air GNSS counterfeit transmission is required for the first four stages, and none was performed for this study.

In three dimensions, the state expands to three translations and three rotation components. At least three non-collinear audited members, plus suitable anchor geometry, are then required generically. Extending SRPL to SE(3), incorporating authenticated Galileo OSNMA status, and designing a sequential scheduler that chooses the member with the greatest expected information gain are promising follow-on contributions.

IX. CONCLUSION

Formation consistency is not absolute integrity. We proved that a rigid GNSS manipulation preserves every pairwise range and can also preserve the position of any permanently audited sentinel. Randomly moving the absolute audit across authenticated member-anchor pairs removes that deterministic blind spot. The resulting information matrix does more than raise an alarm: it states whether the shared translation and yaw are observable, produces an explicit infinite value when they are not, and supports a gated swarm-wide rigid correction. Seeded simulations demonstrate the expected separation among local, common-translation, fixed-sentinel rotation, and slow-drift outcomes while using 96% fewer anchor ranges than continuous polling. The publishable next step is hardware-in-the-loop validation of the detection distribution and SRPL coverage, not a claim that the present simulation is flight-certified.

REPRODUCIBILITY STATEMENT

The source bundle includes the LaTeX manuscript, bibliography, seeded simulation program, raw trial summaries, and vector figures. Parameters used in every reported result are listed in the included manifest.

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